Fire-Aware Robot Dogs

Machine-Learning Wildfire Forecasts for Safe Navigation and Scouting

Shawn Zeng · Monta Vista High School

ONGOING WORK · CURRENTLY IN SIMULATION

built & tested in simulation

1 Predict

The model forecasts **when** each 50 m cell will catch fire. Designed to run on the dog or its controller.

In: satellite fire map, weather, terrain, land cover. Out: **ignition time** for every cell, in 15-min steps.

2 Safe route

Plan paths that avoid cells predicted to ignite **before the dog gets through**.

Fastest-path search: **1.5 m/s** on open ground, 0.7× in forest, 0.5× in brush; slower uphill than downhill; no slopes over 35°. Every cell is crossed at least **30 min** before the forecast says it ignites; the route is re-checked every step.

3 Escape path

Always keep a way out that the forecast says **will stay clear**.

A path home that aims for a stricter **60 min** margin (30 min if it must), re-planned from wherever the dog is. If the way forward closes, the dog retreats along it.

4 Scout

Walk to the fire edge, measure the real conditions, and **get home before the fire arrives**.

Holds **30 min** at the edge (simulated sensors), and leaves early if the fire gets within 30 min of it.

5 Feedback

Send back measured wind, temperature, humidity and the real fire edge so **the next forecast is better**.

The forecast is re-run from the measured conditions and the true fire position; the dog re-plans on it. Demo: match **61%** → **80%**.



robot dog

One simulated mission – from the demo video (QR code, right panel)

Launch
Route and escape path planned on the forecast. Match 82%.

Wind shifts
The true fire turns; the forecast doesn't know. Match drops to **61%**.

Measure
Near the fire, the dog measures wind, temperature, humidity, fire edge.

Report
Forecast re-run on the dog's data. Match back to **80%**.

Re-plan
New route and escape path on the corrected forecast.

Home safe
Back at staging before the fire arrives.

Know where the fire *will* be – not just where it is.

Safer crews
Fewer firefighters sent into danger to look.

Robot stays safe
Routes and exits keep it from getting trapped.

Smarter sensing
Measure where it changes decisions.